

PostgreSQL 18 DBA

Ecosystem, Extensions, Products & Third-Party Tools

End-to-End Interview Cheat Sheet

Companion Guide to PostgreSQL Core Terminology

Technical Baseline: PostgreSQL 18 ecosystem concepts and official project documentation

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Table of Contents

1. Core vs Extension vs Third-Party Tool
2. Connection Pooling & Connectivity
3. High Availability & Failover
4. Distributed Consensus / DCS
5. Load Balancing & Virtual IP
6. Backup & Recovery
7. Monitoring, Metrics & Administration
8. PostgreSQL Extensions
9. Migration & Data Movement Tools
10. Managed Cloud PostgreSQL Products
11. Kubernetes PostgreSQL Operators
12. Popular Production Architecture Stack
13. Interview Classification Master Table
14. Product-to-Purpose Memory Map
15. 30-Second Interview Answer
16. Official / Project References

1. Core vs Extension vs Third-Party Tool

Use this distinction first in interviews. PostgreSQL itself is the database engine. Extensions add capabilities inside PostgreSQL. Third-party products run separately and provide services such as pooling, HA orchestration, backup management, monitoring, migration, or network routing.

PostgreSQL Core

= Database engine + built-in features + supplied utilities

PostgreSQL Extension

= Functionality enabled inside PostgreSQL, commonly with `CREATE EXTENSION`

Third-Party Tool / Product

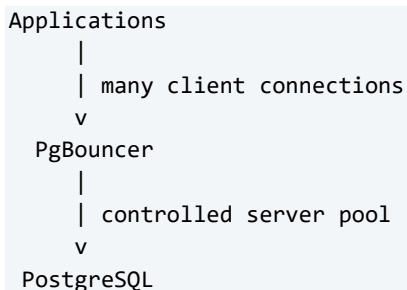
= Separate software that integrates with PostgreSQL

Tool / Product	Category	Core?	Interview-ready purpose
PostgreSQL	Database engine	Yes	Core database server: SQL, MVCC, WAL, transactions, recovery, replication and administration.
pg_stat_statements	Supplied extension	Extension	Aggregated SQL planning/execution statistics; must be configured and enabled.
postgres_fdw	Supplied extension	Extension	Foreign-data access to another PostgreSQL server.
PgBouncer	External tool	No	Connection pooling.
Patroni	External tool	No	PostgreSQL HA orchestration and automated failover.
pgBackRest	External tool	No	Backup, restore and WAL archive management.

2. Connection Pooling & Connectivity

Tool / Product	Category	Core?	Interview-ready purpose
PgBouncer	Connection pooler	No	Lightweight pooler that reuses PostgreSQL server connections and reduces backend connection pressure.
Pgpool-II	Middleware / pooler	No	Pooling middleware with capabilities such as load balancing and additional HA-related functions.
Odyssey	Connection pooler	No	Alternative PostgreSQL connection pooler designed for high connection workloads.

PgBouncer — Interview Focus



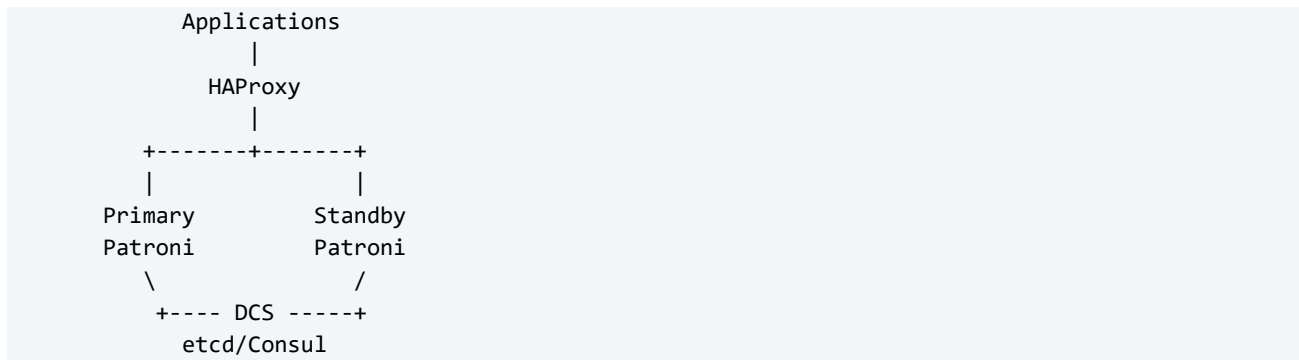
Key terms: Session Pooling, Transaction Pooling, Statement Pooling, Client Connection, Server Connection and Pool Size.

Interview answer: PgBouncer is not PostgreSQL core. It is an external connection pooler used to reduce connection overhead and protect PostgreSQL from excessive backend connections.

3. High Availability & Failover

Tool / Product	Category	Core?	Interview-ready purpose
Patroni	HA orchestration	No	Coordinates PostgreSQL leader election, promotion, failover and switchover using PostgreSQL replication plus a DCS/Kubernetes.
repmgr	Replication / HA management	No	Helps register, monitor, clone, promote and manage PostgreSQL primary/standby nodes; repmgrd can assist with automated failover.

Patroni — Interview Focus



Key terms: Leader, Replica, DCS, Failover, Switchover, patronictl, REST API and synchronous mode.

Interview trap: Patroni does not replace PostgreSQL streaming replication. PostgreSQL replicates the database; Patroni orchestrates node roles and failover.

4. Distributed Consensus / DCS

Tool / Product	Category	Core?	Interview-ready purpose
etcd	Distributed key-value store	No	Frequently used as a Patroni Distributed Configuration Store for cluster coordination.
Consul	Service discovery / KV	No	Can provide distributed coordination and configuration storage for Patroni.
Apache ZooKeeper	Coordination service	No	Distributed coordination service supported by Patroni architectures.

PostgreSQL Streaming Replication
= database/WAL replication

etcd / Consul / ZooKeeper
= cluster coordination / distributed state

Interview trap: etcd does not replicate PostgreSQL data.

5. Load Balancing & Virtual IP

Tool / Product	Category	Core?	Interview-ready purpose
HAProxy	Load balancer / proxy	No	Provides a stable endpoint and can route connections according to PostgreSQL/Patroni node health and role.
Keepalived	VIP / VRRP	No	Provides virtual IP failover and endpoint availability, commonly in front of HAProxy nodes.

```

Application
|
Virtual IP
|
Keepalived
|
HAProxy
|
+-- PostgreSQL Primary
+-- PostgreSQL Replica(s)

```

Interview trap: Keepalived provides network endpoint availability; it does not replicate PostgreSQL or determine database consistency.

6. Backup & Recovery

Tool / Product	Category	Core?	Interview-ready purpose
pgBackRest	Backup / recovery	No	PostgreSQL-focused backup, restore and WAL archive management with full, differential and incremental backup capabilities.
Barman	Backup / recovery	No	Backup and Recovery Manager for PostgreSQL; supports backup, WAL management, retention and recovery workflows.

pgBackRest — Interview Focus

Important terms: Stanza, Repository, Full Backup, Differential Backup, Incremental Backup, Archive Push, Archive Get, Retention and Restore.

```

PostgreSQL
| \
|  +---- WAL ----+
|                    |
+---- Backup --> pgBackRest --> Repository

```

Core distinction: PostgreSQL provides native `pg_basebackup`, `pg_dump/pg_restore`, WAL archiving and PITR capabilities. `pgBackRest` and `Barman` are separate backup-management solutions.

7. Monitoring, Metrics & Administration

Tool / Product	Category	Core?	Interview-ready purpose
pgAdmin	Administration GUI	No	Graphical PostgreSQL administration and development platform.
pgBadger	Log analyzer	No	Analyzes PostgreSQL logs and produces performance/activity

Tool / Product	Category	Core?	Interview-ready purpose
			reports.
Prometheus	Metrics platform	No	Collects and stores time-series metrics; not PostgreSQL-specific.
postgres_exporter	Metrics exporter	No	Exposes PostgreSQL metrics in Prometheus format.
Grafana	Visualization	No	Builds dashboards and visualizes metrics from data sources such as Prometheus.

```

PostgreSQL
|
postgres_exporter
|
Prometheus
|
Grafana
|
Dashboards

```

Interview trap: Grafana visualizes data, Prometheus collects/stores/queries metrics, and an exporter exposes PostgreSQL metrics.

8. PostgreSQL Extensions

Tool / Product	Category	Core?	Interview-ready purpose
pg_stat_statements	Supplied extension	Extension	Aggregated SQL planning/execution statistics for workload analysis.
PostGIS	Extension	No	Adds geospatial types, functions, spatial indexes and related GIS capabilities.
pgvector	Extension	No	Adds vector data types and exact/approximate vector similarity search for AI/embedding workloads.
Citus	Extension / distributed PG	No	Adds distributed tables, sharding and distributed query execution to PostgreSQL.
TimescaleDB	Extension / platform	No	Adds time-series-oriented capabilities such as hypertables and continuous aggregates.
pg_cron	Extension	No	Adds cron-style job scheduling inside PostgreSQL.
pgAudit	Extension	No	Adds detailed audit logging capabilities through PostgreSQL logging.
pg_repack	Extension / tool	No	Reorganizes tables/indexes with reduced locking compared with VACUUM FULL-style rewrites.
pg_partman	Extension	No	Automates management of native PostgreSQL partition sets.
postgres_fdw	Supplied extension	Extension	Accesses remote PostgreSQL tables through the Foreign Data Wrapper interface.
dblink	Supplied extension	Extension	Provides functions for connecting to and querying other PostgreSQL databases.

Common Extension Interview Notes

```
CREATE EXTENSION pg_stat_statements;
CREATE EXTENSION postgres_fdw;
CREATE EXTENSION postgis;
CREATE EXTENSION vector;
```

Do not treat every extension the same. Some extensions are supplied with PostgreSQL distributions, while others such as PostGIS and pgvector are maintained as separate projects.

9. Migration & Data Movement Tools

Tool / Product	Category	Core?	Interview-ready purpose
pgloader	Migration / loading	No	Loads and migrates data into PostgreSQL, including migrations from selected non-PostgreSQL sources.
Ora2Pg	Migration / assessment	No	Assists Oracle-to-PostgreSQL assessment and migration.
AWS DMS	Managed migration service	No	AWS service for homogeneous/heterogeneous database migration and change data capture.
Azure Database Migration Service	Managed migration service	No	Microsoft Azure database migration service.
EDB Migration Toolkit	Commercial ecosystem tool	No	Migration tooling in the EDB PostgreSQL ecosystem.

Interview point: Migration services and conversion tools are not PostgreSQL database-engine features.

10. Managed Cloud PostgreSQL Products

Tool / Product	Category	Core?	Interview-ready purpose
Amazon RDS for PostgreSQL	AWS managed service	No	Managed community-PostgreSQL-based service with AWS-controlled infrastructure and operational features.
Amazon Aurora PostgreSQL-Compatible	AWS database engine	No	PostgreSQL-compatible AWS cloud database engine; architecture is not identical to community PostgreSQL.
Azure Database for PostgreSQL Flexible Server	Azure managed service	No	Microsoft-managed PostgreSQL service.
Cloud SQL for PostgreSQL	Google Cloud managed service	No	Google-managed PostgreSQL service.
AlloyDB for PostgreSQL	Google Cloud platform	No	PostgreSQL-compatible managed database platform with Google-specific architecture/features.
EDB Postgres offerings	Commercial platform	No	Commercial PostgreSQL products, tooling and enterprise services from EDB.
Crunchy Postgres offerings	Commercial / Kubernetes	No	PostgreSQL platforms and Kubernetes-focused tooling from Crunchy Data.

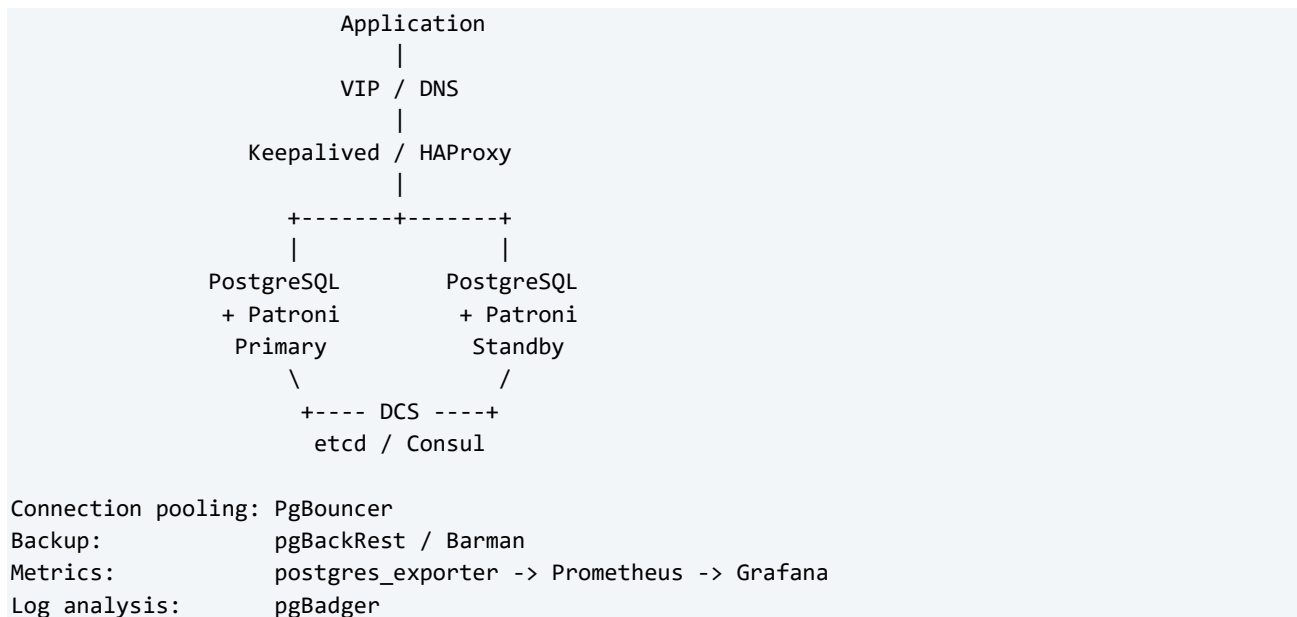
Interview trap: Community PostgreSQL, RDS PostgreSQL, Aurora PostgreSQL-Compatible and Azure Database for PostgreSQL are not operationally identical. Managed services can restrict superuser access, filesystem access, extensions, configuration and upgrade choices.

11. Kubernetes PostgreSQL Operators

Tool / Product	Category	Core?	Interview-ready purpose
CloudNativePG	Kubernetes operator	No	Automates PostgreSQL lifecycle operations in Kubernetes.
Crunchy Postgres Operator	Kubernetes operator	No	Kubernetes-based PostgreSQL provisioning and lifecycle management.
Zalando Postgres Operator	Kubernetes operator	No	Manages PostgreSQL clusters in Kubernetes.

Typical operator responsibilities: provisioning, configuration, replication, failover, backup integration, scaling and Kubernetes lifecycle management.

12. Popular Production Architecture Stack



Interview point: PostgreSQL remains the database engine. The surrounding tools each solve a specific operational problem.

13. Interview Classification Master Table

Tool / Product	Category	Core?	Interview-ready purpose
PostgreSQL	Database engine	Yes	Core database engine.
Streaming Replication	Core feature	Yes	Physical WAL-based replication.
Logical Replication	Core feature	Yes	Publication/subscription logical replication.
Autovacuum	Core feature	Yes	Automatic VACUUM/ANALYZE maintenance.
pg_dump	Core utility	Yes	Logical database backup.
pg_basebackup	Core utility	Yes	Physical base backup.

Tool / Product	Category	Core?	Interview-ready purpose
pg_stat_activity	Core view	Yes	Session/activity monitoring.
pg_stat_statements	Supplied extension	Extension	SQL workload statistics.
postgres_fdw	Supplied extension	Extension	Remote PostgreSQL foreign tables.
PgBouncer	External	No	Connection pooling.
Patroni	External	No	HA orchestration.
HAProxy	External	No	Routing/load balancing.
Keepalived	External	No	VIP/VRP.
etcd	External	No	Distributed coordination.
repmgr	External	No	Replication/HA management.
pgBackRest	External	No	Backup/recovery.
Barman	External	No	Backup/recovery.
pgBadger	External	No	Log analysis.
Prometheus	External	No	Metrics platform.
Grafana	External	No	Dashboards.
PostGIS	Extension	No	Geospatial.
pgvector	Extension	No	Vector similarity search.
Citrus	Extension	No	Distributed/sharded PostgreSQL.
TimescaleDB	Extension/platform	No	Time-series.
pg_cron	Extension	No	Scheduling.
pgAudit	Extension	No	Auditing.
pg_repack	Extension/tool	No	Online-style reorganization.
pg_partman	Extension	No	Partition management.

14. Product-to-Purpose Memory Map

CONNECTIONS	PgBouncer Pgpool-II Odyssey
HA ORCHESTRATION	Patroni repmgr
DCS / CONSENSUS	etcd Consul ZooKeeper
LOAD BALANCING	HAProxy
VIRTUAL IP	Keepalived
BACKUP	pgBackRest Barman
MONITORING	Prometheus postgres_exporter Grafana pgBadger
ADMIN GUI	pgAdmin
AUDITING	pgAudit
BLOAT / REORG	pg_repack
PARTITION AUTO	pg_partman
SCHEDULING	pg_cron pgAgent
GEOSPATIAL	PostGIS
AI / VECTOR	pgvector

SHARDING	Citus
TIME SERIES	TimescaleDB

15. 30-Second Interview Answer

“PostgreSQL provides the core database engine, WAL, MVCC, physical and logical replication, VACUUM, PITR and native administration utilities. Around PostgreSQL, we can use PgBouncer for connection pooling, Patroni with etcd or another DCS for HA orchestration, HAProxy for connection routing, Keepalived for VIP availability, pgBackRest or Barman for backup and recovery, and Prometheus with postgres_exporter and Grafana for monitoring. Extensions such as pg_stat_statements, PostGIS, pgvector, pg_cron, pgAudit, pg_repack and pg_partman add specialized capabilities, while Citus adds distributed/sharded PostgreSQL functionality.”

16. Official / Project References

Project / Product	Official reference
PostgreSQL 18 Documentation	https://www.postgresql.org/docs/18/
PgBouncer	https://www.pgбouncer.org/
Pgpool-II	https://www.pgpool.net/
Patroni	https://patroni.readthedocs.io/
repmgr	https://www.repmgr.org/
etcd	https://etcd.io/
Consul	https://developer.hashicorp.com/consul
Apache ZooKeeper	https://zookeeper.apache.org/
HAProxy	https://www.haproxy.org/
Keepalived	https://www.keepalived.org/
pgBackRest	https://pgbackrest.org/
Barman	https://www.pgbarman.org/
pgAdmin	https://www.pgadmin.org/
pgBadger	https://pgbadger.darold.net/
Prometheus	https://prometheus.io/
Grafana	https://grafana.com/
PostGIS	https://postgis.net/
pgvector	https://github.com/pgvector/pgvector
Citus	https://www.citusdata.com/
Timescale	https://www.timescale.com/
pg_cron	https://github.com/citusdata/pg_cron
pgAudit	https://github.com/pgaudit/pgaudit
pg_repack	https://github.com/reorg/pg_repack
pg_partman	https://github.com/pgpartman/pg_partman